SET 1

Online Grocery Delivery Application

Agile Overview • Scrum Roles, Events and Artifacts • User Stories, Backlog and Estimation

Submitted by

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Project: Online Grocery Delivery | Jira project key: OGD

1. Business Scenario and Product Vision

A regional supermarket chain is launching a grocery ordering and home-delivery application. The business needs a prioritised and estimated product backlog before development begins. Customers should be able to discover grocery products, manage a cart, place and pay for orders, and follow delivery progress.

Product vision statement

For customers who want a convenient way to buy everyday groceries, the Online Grocery Delivery Application is a digital shopping platform that enables customers to browse and search products, check availability, manage a shopping cart, complete secure checkout, and track deliveries. Unlike visiting a physical store or relying on manual ordering, the application provides a simple, transparent and convenient end-to-end grocery shopping experience.

2. Scrum Roles and Anti-patterns

Scrum role	Main responsibility in this project	One anti-pattern to avoid
Product Owner	Owns and communicates the product goal; orders the backlog; clarifies requirements and acceptance criteria; makes value-based priority decisions.	Acting as a task-assigning manager or changing priorities constantly without explaining the product value.
Scrum Master	Ensures Scrum is understood and followed; facilitates events; coaches the team; helps remove impediments.	Behaving like a traditional project manager who commands the Developers or uses the Daily Scrum as a status-report meeting.
Developers	Create a usable Increment; plan and carry out the work; estimate stories; maintain quality and meet the Definition of Done.	Working in isolated silos, avoiding collaboration, or treating testing and quality as someone else's responsibility.

3. Product Backlog: Epics and User Stories

The backlog is organised into three epics. Each user story follows the “As a …, I want …, so that …” format. The Jira project uses the key OGD.

Jira ID	Story	Epic	User story	MoSCoW	Points
OGD-4	Browse Grocery Categories	Browse and Search Products	As a customer, I want to browse grocery categories, so that I can find products by type.	Must	3

Jira ID	Story	Epic	User story	MoSCoW	Points
OGD-5	Search Grocery Products	Browse and Search Products	As a customer, I want to search for grocery products, so that I can quickly find the items I need.	Must	5
OGD-6	View Product Details	Browse and Search Products	As a customer, I want to view product details, so that I can make an informed purchase decision.	Should	3
OGD-7	View Product Availability	Browse and Search Products	As a customer, I want to check product availability, so that I know whether an item can be ordered before adding it to my cart.	Must	2
OGD-8	Add Products to Cart	Cart and Checkout	As a customer, I want to add grocery products to my shopping cart, so that I can purchase multiple items together.	Must	5
OGD-9	Update Cart Quantity	Cart and Checkout	As a customer, I want to increase or decrease product quantities in my cart, so that I can adjust my order.	Should	2
OGD-10	View Cart Total	Cart and Checkout	As a customer, I want to view the total price of my cart, so that I know the amount payable before checkout.	Must	3
OGD-11	Complete Checkout	Cart and Checkout	As a customer, I want to enter my delivery details and review my order, so that I can confirm the purchase.	Must	8
OGD-12	Place Grocery Order	Order and Delivery Tracking	As a customer, I want to place my grocery order, so that the supermarket can process and deliver it.	Must	5

Jira ID	Story	Epic	User story	MoSCoW	Points
OGD-13	Make Secure Payment	Order and Delivery Tracking	As a customer, I want to pay securely for my grocery order, so that my payment information and transaction are protected.	Must	8
OGD-14	View Order History	Order and Delivery Tracking	As a customer, I want to view my previous orders, so that I can review past purchases.	Could	3
OGD-15	Track Order Delivery	Order and Delivery Tracking	As a customer, I want to track my order delivery status, so that I know the progress of my order.	Should	5

4. INVEST Checklist

INVEST is used to check whether a user story is suitable for sprint planning: Independent, Negotiable, Valuable, Estimable, Small and Testable.

INVEST element	What it means	How it is applied to this backlog
Independent	A story should have as few dependencies on other stories as practical.	Stories are grouped by feature area; dependencies such as checkout requiring cart contents are identified during refinement.
Negotiable	A story describes an outcome, not a fixed technical solution.	The stories state customer needs and leave implementation choices to the Developers.
Valuable	A story delivers value to a user or stakeholder.	Each story explains why the customer needs the capability.
Estimable	The team has enough information to estimate the effort.	Descriptions, acceptance criteria and known dependencies are clarified before estimation.
Small	A story is small enough to complete within a sprint.	Large end-to-end capabilities are split into focused stories such as cart quantity, cart total and checkout.
Testable	There are clear conditions for verifying completion.	Acceptance criteria are written in observable Given/When/Then form.

Examples of story refinement

The following refinements make broad stories more specific and testable while preserving the customer outcome.

Original broad story 1: As a customer, I want to manage my cart, so that I can buy groceries.

Refined story 1: As a customer, I want to increase or decrease the quantity of an item already in my cart, so that I can adjust the number of units before checkout.

Original broad story 2: As a customer, I want delivery, so that I receive my groceries.

Refined story 2: As a customer, I want to view the current status of my placed order, so that I can understand whether it is being prepared, dispatched or delivered.

These refinements improve Small and Testable: each describes one customer outcome that can be estimated and verified independently. Dependencies with the cart and order-placement workflow remain visible.

5. Acceptance Criteria (Given / When / Then)

Acceptance criteria define observable behaviour that must be satisfied for a story to be accepted. Five stories were documented, with a negative scenario included for product search.

Story	Acceptance criterion
OGD-4 – Browse Grocery Categories	Given the customer is on the home page, When the customer selects a grocery category, Then the application displays products belonging to that category.
OGD-5 – Search Grocery Products (positive)	Given the customer is on the product search page, When the customer enters a valid product name and submits the search, Then matching products are displayed.
OGD-5 – Search Grocery Products (negative)	Given the customer enters a product name that has no matching items, When the search is submitted, Then the application displays a clear “No products found” message and does not show unrelated products.
OGD-8 – Add Products to Cart	Given a product is available, When the customer selects Add to Cart, Then the product is added to the cart and the cart quantity/count is updated.
OGD-11 – Complete Checkout	Given the cart contains at least one item, When the customer provides valid delivery details and confirms checkout, Then the application displays an order summary for confirmation.
OGD-13 – Make Secure Payment	Given an order is ready for payment, When the customer completes a valid payment, Then the transaction result is recorded and the customer receives a payment confirmation.

6. MoSCoW Prioritisation

MoSCoW classifies requirements by delivery importance: Must have, Should have, Could have and Won't have in this release. The backlog uses Must, Should and Could priorities.

Priority	Meaning	Stories
Must have	Essential for the core grocery shopping and ordering journey.	OGD-4, OGD-5, OGD-7, OGD-8, OGD-10, OGD-11, OGD-12, OGD-13
Should have	Important and valuable, but the product can operate temporarily without it.	OGD-6, OGD-9, OGD-15
Could have	Useful enhancement that can be deferred if time or capacity is limited.	OGD-14
Won't have (this release)	Explicitly excluded from the current release scope.	No story was assigned to this category in the current 12-story backlog.

7. Fibonacci Story-Point Estimation and Planning Poker

The team uses relative estimation with Fibonacci values (1, 2, 3, 5, 8, 13, …). Story points represent relative effort, complexity and uncertainty, not hours. Planning Poker allows team members to estimate independently, discuss differences and agree on a value.

Reference point	Interpretation
2 points	A small, straightforward change with limited complexity; used as a low-complexity comparison.
3 points	A modest feature requiring some validation and integration.
5 points	A medium-sized story with multiple steps or meaningful business rules.
8 points	A larger or more uncertain workflow involving several validations or system interactions.

Reference-story justification: OGD-4 (Browse Grocery Categories) is treated as a 3-point reference. It represents a relatively clear, moderate feature: displaying categories and navigating to the selected category. Other stories are compared against this baseline. For example, OGD-7 is smaller (2 points), while OGD-11 and OGD-13 are larger (8 points) because checkout and payment involve more steps, validation and failure handling.

Jira ID	Story	Points
OGD-4	Browse Grocery Categories	3
OGD-5	Search Grocery Products	5
OGD-6	View Product Details	3
OGD-7	View Product Availability	2
OGD-8	Add Products to Cart	5
OGD-9	Update Cart Quantity	2
OGD-10	View Cart Total	3
OGD-11	Complete Checkout	8
OGD-12	Place Grocery Order	5
OGD-13	Make Secure Payment	8
OGD-14	View Order History	3
OGD-15	Track Order Delivery	5

Total backlog estimate: 52 story points

8. Sprint Planning

The 12 stories are distributed across three one-week sprints. Each sprint has a clear feature goal and a seven-day date range.

Sprint	Dates	Sprint goal	Stories	Total points
OGD Sprint 1	10 Nov – 17 Nov	Complete grocery category browsing, product search, product details and availability.	OGD-4, OGD-5, OGD-6, OGD-7	13
OGD Sprint 2	17 Nov – 24 Nov	Deliver cart management and checkout capabilities.	OGD-8, OGD-9, OGD-10, OGD-11	18
OGD Sprint 3	24 Nov – 1 Dec	Deliver order placement, secure payment, order history and delivery tracking.	OGD-12, OGD-13, OGD-14, OGD-15	21

The planned estimates are 13, 18 and 21 points respectively. These are planning allocations, not proof that the work has been completed. Jira status fields should reflect the actual progress of each story.

9. Definition of Ready (DoR) and Definition of Done (DoD)

Definition of Ready

A user story is ready for development when:

- Requirements are clearly documented.
- Acceptance criteria are defined.
- The user story is understandable to the development team.
- Dependencies are identified.
- Required designs or mockups are available.
- Story points are estimated.
- The Product Owner has approved the requirements.

Definition of Done

A user story is considered complete when:

- Development is completed.
- Code follows project standards.
- Unit testing is completed.
- Integration testing is completed.
- No critical bugs remain.
- Acceptance criteria are satisfied.
- Code is reviewed and approved.
- The feature is successfully integrated.

Project scope note

The Confluence page also includes a Project Scope section. The DoR and DoD provide shared quality expectations for backlog refinement, sprint execution and acceptance.

10. Jira and Confluence Implementation Evidence

The project was configured in Jira as the Online Grocery Delivery Scrum project (OGD). Three epics were created and the 12 user stories were added to the backlog with story points, priority and sprint allocation. The published Confluence page records the Definition of Ready and Definition of Done.

Confluence evidence – Definition of Ready

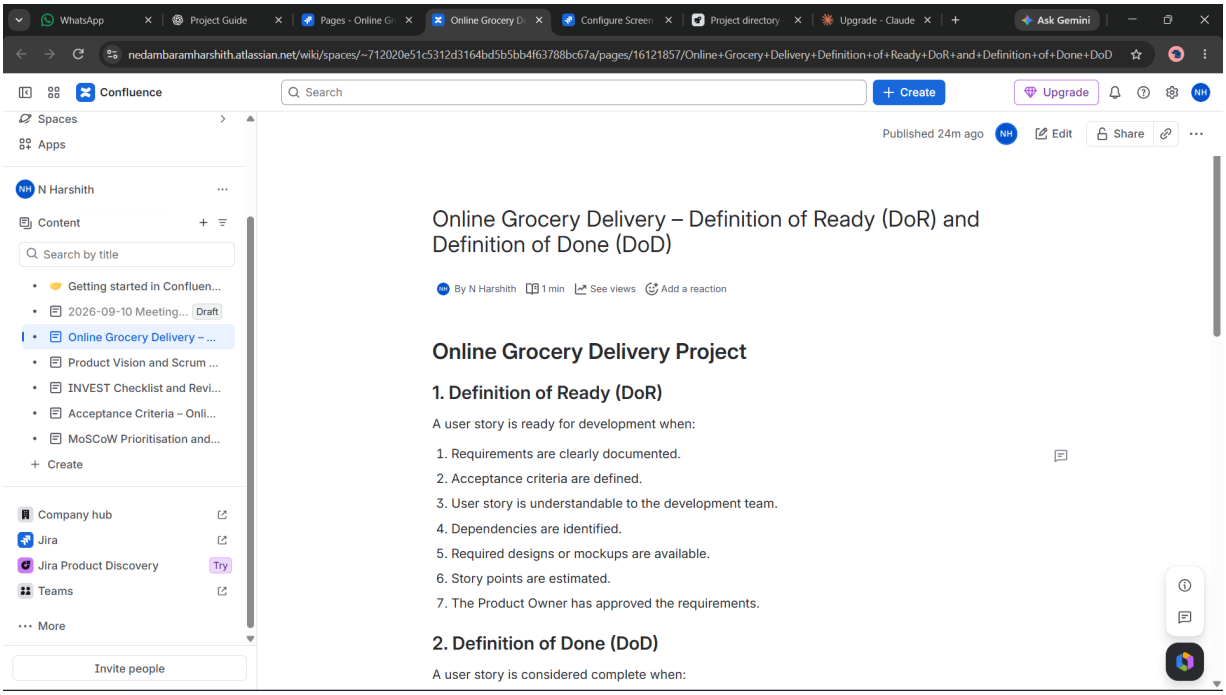


Figure 1. Published Confluence page showing the DoR and the beginning of the DoD.

Confluence evidence – Definition of Done

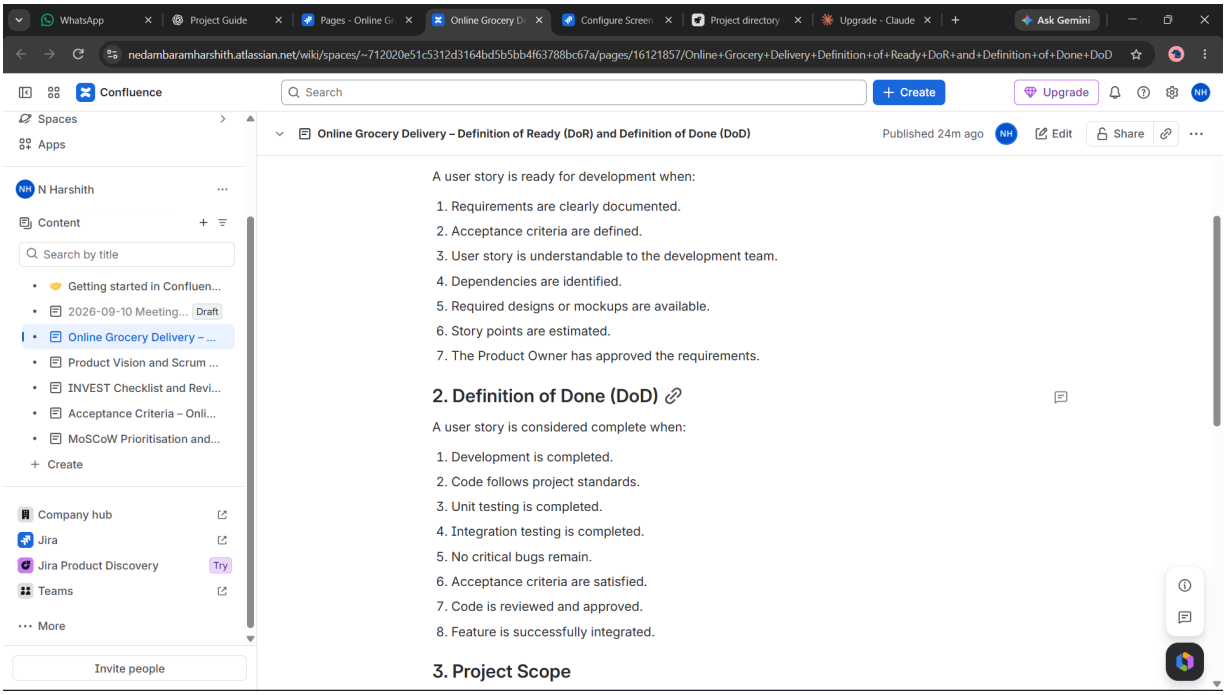


Figure 2. Published Confluence page showing the DoD checklist.

11. Conclusion

The Online Grocery Delivery project has a defined product vision, Scrum role responsibilities, three epics and a 12-story prioritised backlog. The stories have been assessed using INVEST, acceptance criteria have been prepared for five stories, and Fibonacci story points provide a relative estimate of the work. The backlog is allocated across three one-week sprints, totalling 52 story points. The published DoR and DoD establish shared expectations for when work can begin and when a story can be considered complete.

12. Jira Board Evidence – Backlog and Sprint Allocation

The following screenshots provide evidence of the Jira implementation. They show the 12 user stories in the backlog, their epic grouping and story-point estimates, and the allocation of stories across the three one-week sprints.

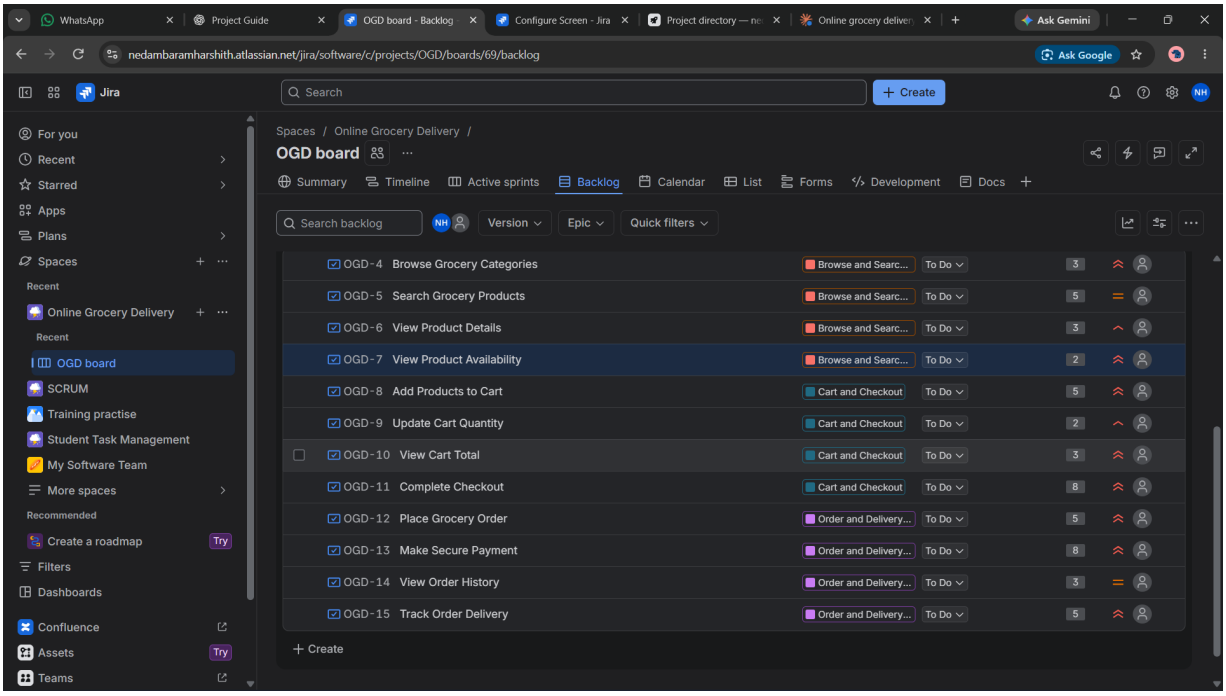


Figure 3. Jira backlog showing all 12 user stories grouped under the three project epics, with story-point estimates.

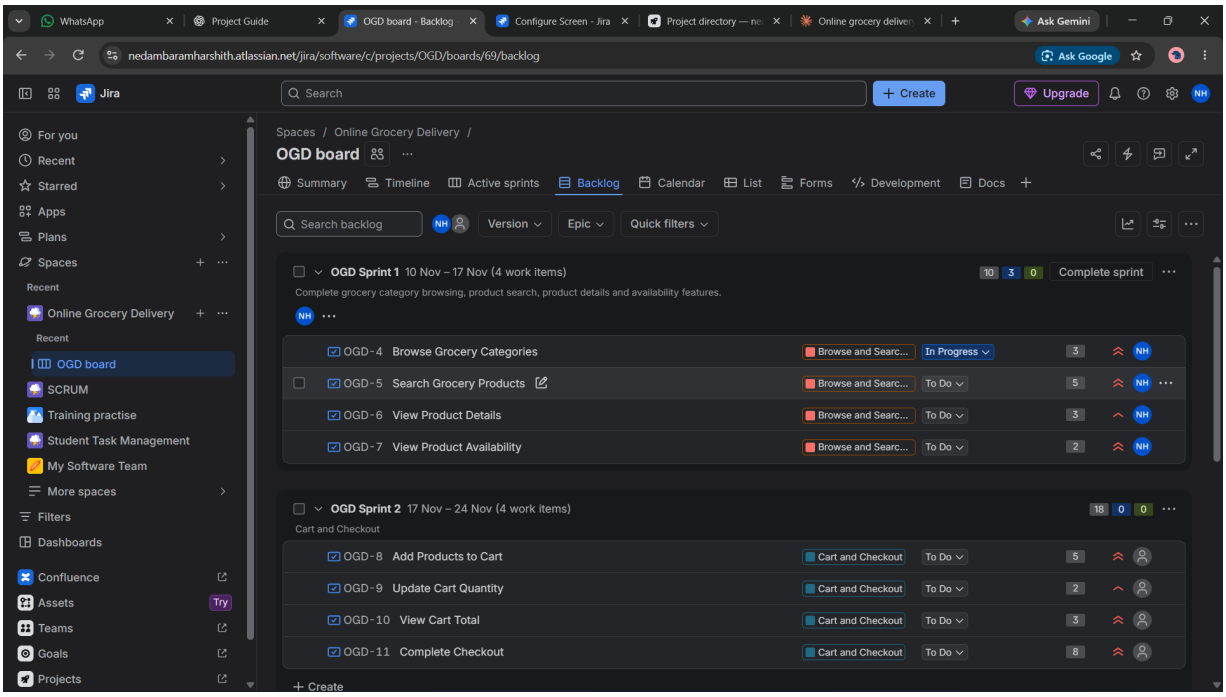


Figure 4. Jira backlog showing Sprint 1 and Sprint 2, their sprint goals, assigned stories and estimates.

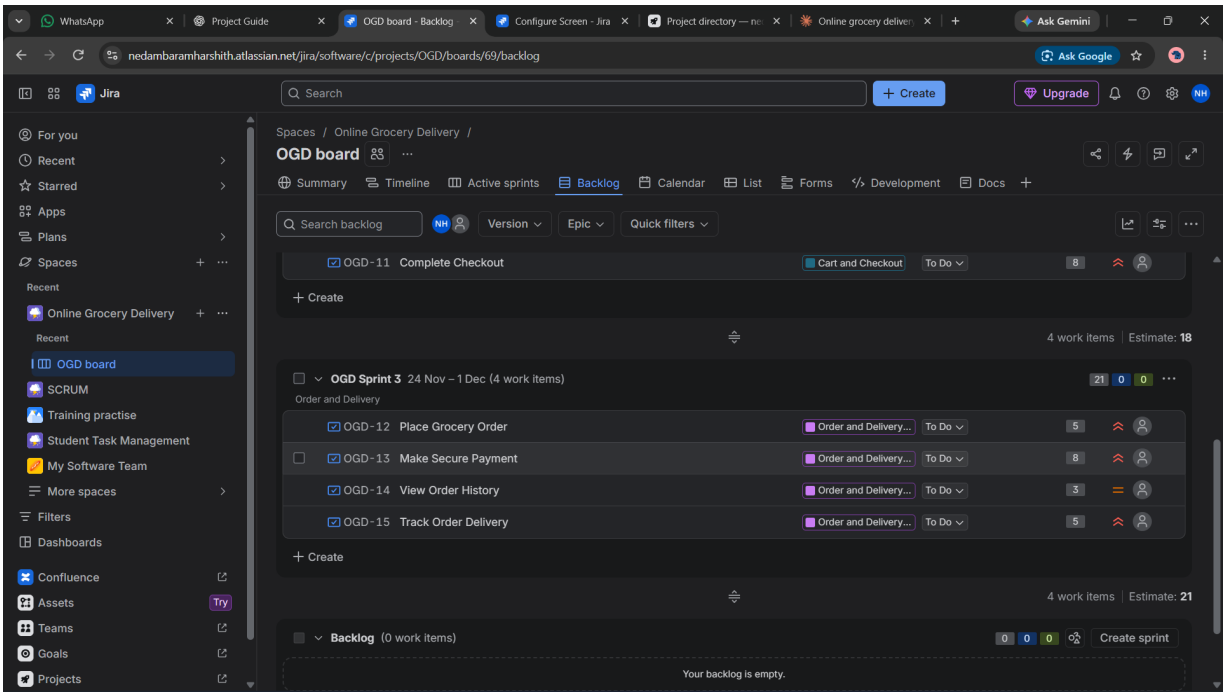


Figure 5. Jira backlog showing Sprint 3 with the order and delivery stories, and an empty remaining backlog.